# 🧬 CBSE Class 10 Science (2026) Premium Notes

## Chapter 6: Life Processes

*Master the fundamentals of biology with these visually engaging, highly detailed, and professionally designed notes.*

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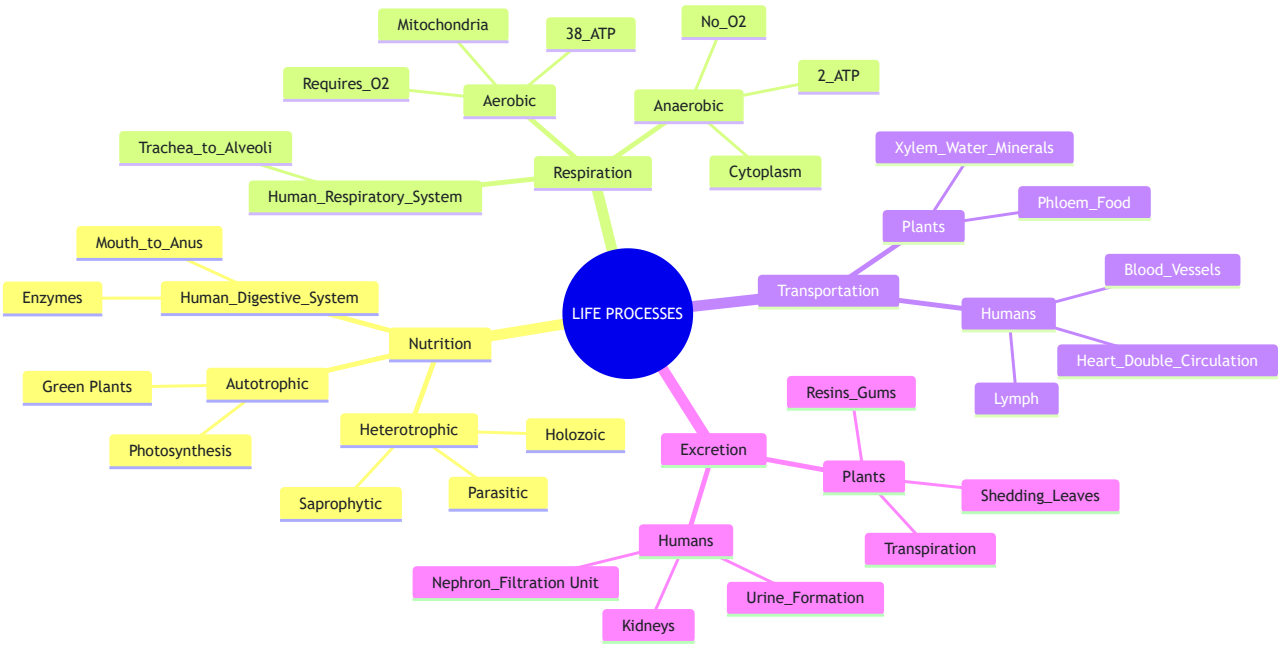
## 🗺️ Master Mind Map

*(Get a bird's-eye view of the entire chapter before diving in!)*

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## 🌟 Introduction to Life Processes

**Definition:** The fundamental, basic processes that are naturally performed by living organisms to maintain their existence on Earth are called **Life Processes**.

Even when an organism is resting or sleeping, these maintenance jobs must continue to prevent damage and breakdown of the body. Since these maintenance processes require energy, the energy must come from outside the body of the individual organism.

**The Four Core Vital Processes:**

1. 🍔 **Nutrition:** The process of taking in food and using it for growth, metabolism, and repair.
2. 🌬️ **Respiration:** The process of breaking down food sources to release energy for cellular needs.
3. 🚚 **Transportation:** The circulation of vital substances (like food, oxygen, water) from one part of the body to another.
4. 🗑️ **Excretion:** The process of safely removing harmful metabolic waste by-products from the body.

> **Important**
> **Energy Transfer:** The energy required for life processes comes from carbon-based molecules (food). Since the environment is not under the control of the individual organism, the outside source of energy is quite varied.

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## 🍔 1. NUTRITION

**Definition:** Nutrition is the process by which an organism procures its nourishment (food) from its surroundings and utilizes it to derive energy for maintenance and growth.

### 📊 Modes of Nutrition

| Mode | Definition | Sub-types | Examples |
|---|---|---|---|
| **Autotrophic** | Organisms synthesize their own organic food from simple inorganic substances ($CO_2$ and $H_2O$) using sunlight. | Photoautotrophic, Chemoautotrophic | Green plants, Cyanobacteria (Blue-green algae) |
| **Heterotrophic** | Organisms cannot synthesize their own food and depend on others for their nutritional requirements. | Holozoic, Saprophytic, Parasitic | Humans, Fungi, Cuscuta, Amoeba |

**Heterotrophic Sub-types in Detail:**

- **Holozoic Nutrition:** Organisms take in solid, complex food materials and break them down internally. *(Example: Humans, Amoeba, Dogs)*
- **Saprophytic Nutrition:** Organisms secrete enzymes outside their body to break down dead, decaying organic matter and then absorb the simple nutrients. *(Example: Bread*

*mould, Mushrooms, Yeast)*

- **Parasitic Nutrition:** Organisms live on or inside the body of another living organism (the host) and derive their food from it, usually without immediately killing the host. *(Example: Cuscuta/Amarbel, Ticks, Lice, Tapeworms)*

## 🌿 Autotrophic Nutrition (Photosynthesis)

**Photosynthesis** is the complex physiological process by which green plants manufacture their own food (glucose) using carbon dioxide and water in the presence of sunlight and chlorophyll.

> 💡 **Tip**
>
> **Key Equation for Photosynthesis:** $$ 6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 \text{ (Glucose)} + 6O_2 + 6H_2O $$

🔑 **Three Major Events in Photosynthesis:**

1. **Absorption:** Trapping of light energy by the green pigment, **chlorophyll**.
2. **Conversion & Splitting:** Conversion of light energy into chemical energy, and the splitting of water molecules ($H_2O$) into Hydrogen and Oxygen using that energy (Photolysis).
3. **Reduction:** The reduction of carbon dioxide ($CO_2$) to form carbohydrates (Glucose).

*(Note: These steps do not always happen one after the other immediately. For instance, desert plants take up $CO_2$ at night and prepare an intermediate compound, which is acted upon by sunlight energy during the day.)*

🔬 **Diagram Description: Stomata & Cross-Section of Leaf**

- **Stomata:** Tiny, microscopic pores present predominantly on the lower epidermis of leaves. They are essential for massive gaseous exchange ($CO_2$ in, $O_2$ out) and transpiration.
- **Guard Cells:** Two kidney-shaped cells surrounding the stomatal pore. They regulate the opening and closing of the pore.
  - *When water flows INTO guard cells:* They swell, become turgid, and the pore **opens**.
  - *When water flows OUT of guard cells:* They shrink, become flaccid, and the pore **closes**.
- **Chloroplasts:** The green dot-like cell organelles found inside mesophyll cells containing chlorophyll, making them the actual site of photosynthesis.

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## 🧑‍🤝‍🧑 Nutrition in Human Beings (Human Digestive System)

The human digestive system is basically a long, continuous tube extending from the mouth to the anus, known as the **Alimentary Canal**, along with associated digestive glands.

> ⓘ **Note**
>
> **The 5 Steps of Holozoic Nutrition:** Ingestion $\rightarrow$ Digestion $\rightarrow$ Absorption $\rightarrow$ Assimilation $\rightarrow$ Egestion.

**Step-by-Step Breakdown of Digestion:**

1. **The Mouth (Buccal Cavity):**
   - **Teeth** crush the food into smaller particles.
   - **Salivary Glands** secrete **Saliva**, containing the enzyme **Salivary Amylase**.
   - *Action:* Salivary amylase breaks down complex starch into simpler sugars (maltose).
2. **Oesophagus (Food Pipe):**
   - Food is pushed down by rhythmic contraction and relaxation of the canal muscles. This movement is called **Peristalsis**.
3. **Stomach:**
   - A large, J-shaped organ that expands when food enters. The muscular walls help in mixing the food thoroughly with gastric juices.
   - **Gastric Glands** secrete three crucial substances:
     - **Hydrochloric Acid (HCl):** Creates an acidic medium, which facilitates the action of the enzyme pepsin. It also kills harmful bacteria in food.
     - **Pepsin (Enzyme):** Active only in an acidic medium; breaks down proteins into peptides.
     - **Mucus:** Protects the inner lining of the stomach from the corrosive action of its own HCl under normal conditions.
4. **Small Intestine (The Site of Complete Digestion):**
   - The longest part of the alimentary canal. It receives secretions from two major glands: the Liver and the Pancreas.
   - **The Liver:** Secrets **Bile Juice** (stored in the gall bladder). Bile is alkaline and contains bile salts. It breaks down large fat globules into smaller miscelles, increasing the efficiency of enzyme action. This is called **Emulsification**.
   - **The Pancreas:** Secretes **Pancreatic Juice** containing enzymes:
     - **Trypsin:** For digesting proteins.
     - **Lipase:** For breaking down emulsified fats.
   - **Intestinal Glands:** Secrete intestinal juice that completes the final conversion:
     - Proteins $\rightarrow$ **Amino Acids**
     - Complex Carbohydrates $\rightarrow$ **Glucose**
     - Fats $\rightarrow$ **Fatty Acids + Glycerol**
   - **Absorption Mechanism:** The inner wall of the small intestine has millions of microscopic, finger-like projections called **Villi**. Villi massively increase the surface

area for rapid absorption of digested food into the deeply supplied blood vessels.

5. **Large Intestine:**
    - Unabsorbed food is sent here. More villi absorb excess water from the undigested material. The rest of the solid waste is removed from the body via the **anus**.

> ⚠ Warning
>
> **Common Mistakes Students Make:**
>
> - *Mistake:* Thinking Digestion starts in the stomach.
> - *Correction:* Digestion of starch begins right in the **mouth** due to salivary amylase!
> - *Mistake:* Stating Bile Juice contains digestive enzymes.
> - *Correction:* Bile juice contains **no enzymes**. Its primary functions are rendering the acidic food from stomach alkaline and emulsifying fats.

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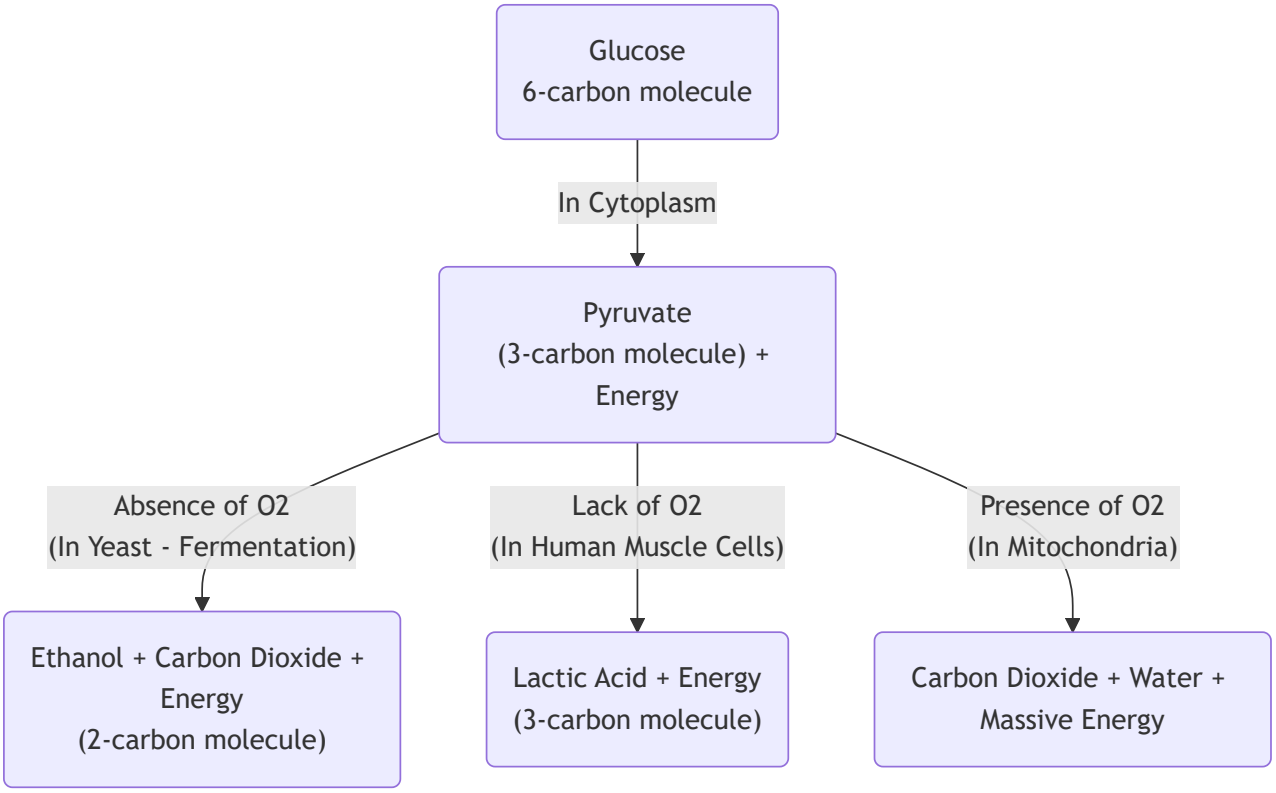
## 🌬️ 2. RESPIRATION

**Definition:** Respiration is a complex bio-chemical process taking place inside cells where food (glucose) is oxidized to release energy. The energy released is immediately stored in the form of **ATP (Adenosine Triphosphate)** molecules.

> Important
>
> **ATP (The Energy Currency of the Cell):** ATP is used to fuel all other endothermic cellular activities. Breaking the terminal phosphate linkage in ATP releases a fixed, large amount of energy (equivalent to 30.5 kJ/mol).

### 🔄 Breakdown of Glucose by Various Pathways

The first step in any respiratory pathway takes place in the **cytoplasm**. Here, one 6-carbon molecule of **Glucose** is broken down into two 3-carbon molecules called **Pyruvate**. The fate of Pyruvate depends entirely on the presence or absence of Oxygen ($O_2$).




Types of Respiration

| Feature | Aerobic Respiration | Anaerobic Respiration |
|---|---|---|
| **Oxygen requirement** | Requires plentiful Oxygen. | Occurs in the absence/lack of Oxygen. |
| **Site of reaction** | Cytoplasm $\rightarrow$ **Mitochondria** | **Cytoplasm** only. |
| **End Products** | $CO_2$, Water, Energy | Ethanol + $CO_2$ (yeast) OR Lactic Acid (muscles) |
| **Energy Yield** | Very High (**38 ATP** per glucose) | Very Low (**2 ATP** per glucose) |
| **Equation** | Glucose + $6O_2 \rightarrow 6CO_2 + 6H_2O$ + 38ATP | Glucose $\rightarrow$ Ethanol + $CO_2$ + 2ATP |

> ⓘ **Note**
>
> **Muscle Cramps Mechanism:** When you exercise vigorously, oxygen demand spikes. If your body cannot supply enough $O_2$ fast enough, muscle cells switch to *anaerobic respiration*. The sudden accumulation of **Lactic Acid** in the muscles is what causes painful cramps.

## 🫁 Human Respiratory System

The pathway of air in human beings: **Nostrils** $\rightarrow$ **Nasal Cavity** (air is warmed, moistened, and filtered by fine hairs/mucus) $\rightarrow$ **Pharynx** $\rightarrow$ **Larynx**

(voice box) $\rightarrow$ **Trachea** (windpipe) $\rightarrow$ **Bronchi** $\rightarrow$ **Bronchioles** $\rightarrow$ **Alveoli**.

- **Rings of Cartilage:** The trachea is supported by rings of cartilage (C-shaped). These rings ensure that the air passage does not collapse even when there is little air in it.

🔬 **Diagram Description: Alveoli Structure & Function**

- **Alveoli:** Within the lungs, the passage divides into smaller tubes ending in tiny, balloon-like structures called alveoli.
- **Why are they crucial?** They provide an immensely vast surface area (if unfolded, about 80 square meters!) for rapid exchange of gases.
- The muscular walls of the alveoli contain an extensive network of incredibly thin blood capillaries. As air is inhaled, Oxygen rapidly diffuses across the thin alveolar wall into the blood (where it binds to the respiratory pigment, **Haemoglobin** in RBCs). Simultaneously, Carbon Dioxide diffuses from the blood into the alveoli to be exhaled.

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# 🩸 3. TRANSPORTATION

## 🫀 Transportation in Human Beings

The human circulatory system is responsible for the transport of materials throughout the body. It consists of three major components:

1. **A Pumping Organ:** The Heart
2. **A Network of Tubes:** Blood Vessels (Arteries, Veins, Capillaries)
3. **The Carrier Mediums:** Blood and Lymph

### The Blood

Blood is a fluid connective tissue.

- **Plasma:** The fluid, yellowish matrix transporting food, $CO_2$, and nitrogenous wastes in a dissolved form.
- **Red Blood Cells (RBCs):** Contain Haemoglobin, which has a very high affinity for oxygen, transporting $O_2$ to all cells.
- **White Blood Cells (WBCs):** The immune army—they fight disease-causing microorganisms.
- **Platelets:** Cell fragments essential for blood clotting. If a vessel gets injured, platelets plug the leak by forming a clot, preventing fatal blood loss and maintaining pressure.

### The Human Heart & Double Circulation

The heart is a muscular organ roughly the size of a clenched fist. Because both $O_2$ and $CO_2$ must be transported by the blood, the heart has four chambers to completely isolate oxygen-rich blood from carbon dioxide-rich blood.

- **Upper Chambers:** Left Atrium & Right Atrium (thin-walled, receiving chambers).
- **Lower Chambers:** Left Ventricle & Right Ventricle (thick, muscular walls, pumping chambers).

> **Tip**
>
> **The Flow of Blood (Double Circulation):** Blood travels through the heart *twice* during one completely cycle.
>
> 1. **Pulmonary Circulation:** Deoxygenated blood from the body enters the **Right Atrium** via Vena Cava $\rightarrow$ passes to **Right Ventricle** $\rightarrow$ pumped to **Lungs** via Pulmonary Artery for oxygenation.
> 2. **Systemic Circulation:** Oxygenated blood returns from Lungs to **Left Atrium** via Pulmonary Vein $\rightarrow$ passes to **Left Ventricle** $\rightarrow$ forcefully pumped to the **rest of the body** via the massive Aorta.

**Blood Vessels: Arteries vs. Veins**

| Characteristic | Arteries | Veins | Capillaries |
|---|---|---|---|
| **Direction of Flow** | AWAY from heart to organs | AWAY from organs to heart | Connect Arteries and Veins |
| **Blood Type** | Usually Oxygenated *(Exception: Pulmonary Artery)* | Usually Deoxygenated *(Exception: Pulmonary Vein)* | Mixed / Exchange site |
| **Wall Structure** | Thick, highly elastic | Thin, less elastic | Single-cell thick walls |
| **Pressure** | High internal pressure | Low internal pressure | Extremely low pressure |
| **Valves** | **Absent** | **Present** (ensures one-way flow back to heart) | Absent |

**The Invisible Liquid: Lymph**

Lymph (tissue fluid) is another transport fluid. It is similar to blood plasma but colorless and contains much less protein. It drains into lymphatic capillaries from intercellular spaces, carrying digested and absorbed fat from the intestine and draining excess fluid back into the blood.

### 🌳 Transportation in Plants

Plants have low energy needs due to a larger proportion of dead cells and a lack of mobility. Therefore, their transport systems can afford to be relatively slow. They use two completely distinct pathways constructed as independently organized conducting tubes:

1. **Xylem (The Water Pipe):**
   - Moves **Water and Minerals** obtained from the soil.
   - Consists of vessels and tracheids.
   - **Direction:** Unidirectional (Only upwards: roots $\rightarrow$ stem $\rightarrow$ leaves).
   - **Mechanism (Transpiration Pull):** The evaporation of water molecules from the cells of a leaf (Transpiration) creates a strong suction force that rapidly pulls water up from the xylem cells of roots, much like sucking water through a straw. It works best during the daytime.
2. **Phloem (The Food Pipe):**
   - Moves **Food materials** (sucrose, amino acids) synthesized in the leaves during photosynthesis.
   - Consists of sieve tubes and companion cells.
   - **Direction:** Bidirectional (Upwards to growing tips AND downwards to storage organs like roots/fruits).
   - **Mechanism (Translocation):** Achieved by utilizing energy. Material like sucrose is loaded into the phloem tissue using energy from **ATP**. This massively increases the osmotic pressure inside the tissue causing water to rapidly rush in, which forcibly moves the material to nearby tissues with lower pressure.

> ⚠ **Warning**
>
> **Common Mistakes Students Make:**
>
> - *Mistake:* Believing both xylem and phloem transport happens via simple physical suction.
> - *Correction:* While xylem heavily relies on physical forces like transpiration pull, phloem transport requires actual active biological energy expenditures via **ATP**!

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## 🚽 4. EXCRETION

**Definition:** The biological process responsible for gathering and removing toxic, harmful metabolic waste products (such as urea, uric acid) from the body is called excretion.

### 👩 Excretion in Human Beings

The human excretory system consists of:

1. **A pair of Kidneys:** Located in the abdomen, one on either side of the backbone.
2. **A pair of Ureters:** Tubes connecting kidneys to the urinary bladder.
3. **A Urinary Bladder:** A muscular sac that stores urine until it is passed out. This is under nervous control.
4. **A Urethra:** The tube through which urine is expelled from the body.

**The Nephron: The Filtration Factory**

The basic filtration unit inside the kidney is a cluster of very thin-walled blood capillaries. Each kidney contains over a million independent, complex tubular structures called **Nephrons**.

🔬 **Diagram Description: Parts of a Nephron**

- **Bowman's Capsule:** A cup-shaped upper end of a nephron. It encapsulates a bundle of tightly coiled blood capillaries called the **Glomerulus**. High-pressure ultrafiltration of blood happens here.
- **Tubular Part:** A long coiled tube. As the initial filtrate passes through this tube, crucial useful substances (like glucose, amino acids, salts, and vast amounts of water) are intentionally **selectively reabsorbed** back into the capillaries surrounding the tubule.
- **Collecting Duct:** Only the true waste (urea, excess water, excess salts) remains, now known as **Urine**, which enters the collecting duct to be channeled to the ureter.

> 💡 Tip
> **Artificial Kidney (Hemodialysis):** When kidneys completely fail due to infection or injury, toxic wastes violently accumulate in the blood. Dialysis acts as an artificial kidney containing long tubes with a semi-permeable lining suspended in dialyzing fluid. Due to concentration gradients, exactly like a real kidney, wastes diffuse out of the patient's blood into the fluid. However, *unlike a real kidney, there is no selective reabsorption*!

## 🌿 Excretion in Plants

Plants use completely different, fascinating strategies for excretion compared to animals:

1. **Gaseous Wastes:** Oxygen ($O_2$) is a direct biological waste product of photosynthesis, released via stomata during the day. Carbon dioxide ($CO_2$) is the waste product of plant respiration, released at night.
2. **Liquid Wastes:** Excess water is simply removed by evaporation in a process known as **Transpiration**.
3. **Solid / Stored Wastes:**
   - Many waste products are intentionally stored in cellular vacuoles.
   - Wastes are often stored in leaves; when the leaf becomes saturated and ages, it simply dramatically falls off the plant (a clever way to dump garbage).

- Wastes are excreted as magnificent **Resins and Gums**, especially in old, dead xylem tissues of the plant.
- Some wastes are also secreted downward directly into the soil surrounding the root zone.

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## ⚡ SUMMARY & QUICK REVISION

- **Autotrophic nutrition** synthesizes complex hydrocarbons via sunlight; **Heterotrophic nutrition** involves the breakdown of complex material acquired from others.
- **Carbohydrates** are digested to glucose; **Proteins** to amino acids; **Fats** to fatty acids and glycerol.
- **Aerobic respiration** creates 38 ATP per glucose molecule exclusively inside mitochondria. **Anaerobic respiration** creates only 2 ATP strictly in the cytoplasm.
- The human heart undergoes **Double Circulation**, completely separating oxygenated and deoxygenated blood to maintain high metabolic rates and body temperature.
- **Arteries** (thick, elastic) carry blood away from the heart; **Veins** (thin, with valves) bring it back. **Capillaries** are the microscopic exchange sites.
- Plant transport relies on **Xylem** (water/minerals moving up due to transpiration pull) and **Phloem** (food moving bidirectionally using ATP energy).
- The **Kidney's Nephron** uses a two-step process: high-pressure filtration at the Glomerulus, followed by selective reabsorption along the tubular portion, creating Urine.

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✨ **You are now ready to ace the "Life Processes" chapter in your CBSE 2026 Science Exam!** ✨

*Practice drawing the Heart, Nephron, Human Digestive System, and Leaf Cross-section.*